

Linear Algebra

Problem Sheet 1

Daniel Greenfield

Question 1.1.

a.

b. let $v_1, v_2, \dots, v_k$ be k arbitrary vectors in $\mathbb{R}^n$. For $k \in \mathbb{Z}_+$ let $P(k)$ be the statement:

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

We will use induction to prove that $\forall k \in \mathbb{Z}_+ P(k)$ is true. For $P(1)$

$$\|v_1\| \leq \|v_1\|$$

is true as the left hand side and the right hand side are equal.

For $P(2)$

$$\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$$

is true due to the triangle inequality ($\|u + v\| \leq \|u\| + \|v\|$). Now assume that $P(K)$ is true for some integers $k > 2$.

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

Adding $\|v_{k+1}\|$ where $v_{k+1} \in \mathbb{R}^n$ is some arbitrary vector in $\mathbb{R}^n$ to both sides will not change the truth value of the inequality giving us

$$\|v_1 + v_2 + \dots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| + \|v_{k+1}\|$$

Using the triangle inequality again we can get a second inequality involving the left hand side:

$$\|v_1 + v_2 + \dots + v_k + v_{k+1}\| \leq \|v_1 + v_2 + \dots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| + \|v_{k+1}\|$$

Due to the transitive properties of inequalities we get that:

$$\|v_1 + v_2 + \dots + v_k + v_{k+1}\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| + \|v_{k+1}\|$$

Which is the statement $P(k+1)$ is so we have shown that if $P(k)$ is true then $P(k+1)$ is true.

So we have shown that for $k > 2$, $P(k) \Rightarrow P(k+1)$ and that $P(1)$ and $P(2)$ are true. Then by the method of induction we have shown that $P(k)$ is true for all $k \in \mathbb{Z}_+$.

Question 1.1.

a. Let $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ y \end{pmatrix}$ with $y \in \mathbb{R}$ then

$$\|u\| = 1$$

$$\|v\| = \sqrt{1 + y^2}$$

$$\|u + v\| = \sqrt{4 + y^2}$$

Finding $y \in \mathbb{R}$ such that $\|u + v\| = \|u\| + \|v\|$.

$$\sqrt{4 + y^2} = 1 + \sqrt{1 + y^2}$$

Squaring both sides gives us

$$4 + y^2 = 1 + 1 + y^2 + 2 \cdot \sqrt{1 + y^2}$$

Which can be simplified to:

$$1 = \sqrt{1 + y^2}$$

Squaring both sides again leaves us with

$$1 = 1 + y^2$$

Therefore $y = 0$.

b. Let $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ be k arbitrary vectors in $\mathbb{R}^n$. Prove that:

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

Proof: Let $\{A_m \in \mathbb{R} : m \in [0, k]\}$ such that A_m is

$$A_m = \|v_1\| + \|v_2\| + \dots + \|v_m\| + \|v_{m+1} + \dots + v_k\| = \sum_{i \in [1, m]} \|v_i\| + \|\sum_{i \in [m+1, k]} v_i\|$$

So that $A_0 = \|v_1 + v_2 + \dots + v_k\|$ and $A_k = \|v_1\| + \|v_2\| + \dots + \|v_k\|$.

For any A_m

$$A_m = \|v_1\| + \|v_2\| + \dots + \|v_{m-1}\| + \|v_m\| + \|v_{m+1} + \dots + v_k\|$$

By utilising the triangle inequality ($\|u + v\| \leq \|u\| + \|v\|$) where $u = v_m$ and $v = v_{m+1} + \dots + v_k$. We get

$$A_m \geq \|v_1\| + \|v_2\| + \dots + \|v_{m-1}\| + \|v_m + (v_{m+1} + \dots + v_k)\|$$

Therefore by definition of A

$$A_m \geq A_{m-1}, \forall m \in [0, k]$$

Giving us the inequality

$$A_0 \leq A_1 \leq \dots \leq A_k$$

Therefore by the transitive property of inequalities we get $A_0 \leq A_k$ and hence

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| \quad \blacksquare$$

Question 1.1. (Alt)

a.

b. Let $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ be k arbitrary vectors in $\mathbb{R}^n$. Proof of:

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

for $k = 2$ the inequality is $\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$ which is just the triangle inequality and therefore true

Assuming that the statement is true for some value k such that

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

If we add $\|v_{k+1}\|$ to both sides we get:

$$\|v_1 + v_2 + \dots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| + \|v_{k+1}\|$$

Due to the triangle inequality ($\|u + v\| \leq \|u\| + \|v\|$) we get this inequality with the left hand side:

$$\|v_1 + v_2 + \dots + v_k + v_{k+1}\| \leq \|v_1 + v_2 + \dots + v_k\| + \|v_{k+1}\|$$

Combining these two inequalities we get

$$\|v_1 + v_2 + \dots + v_k + v_{k+1}\| \leq \|v_1 + v_2 + \dots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| + \|v_{k+1}\|$$

Therefore due to the transitive properties

Question 1.1. (Alt2)

a.

b. let $v_1, v_2, \dots, v_k$ be k arbitrary vectors in $\mathbb{R}^n$. For $k \in \mathbb{Z}_+$ let $P(k)$ be the statement:

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We will use induction to prove that $\forall k \in \mathbb{Z}_+ P(k)$ is true. For $P(1)$

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is true as the left hand side and the right hand side are equal.

For $P(2)$

$$\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$$

is true due to the triangle inequality ($\|u + v\| \leq \|u\| + \|v\|$). Now assume that $P(K)$ is true for some integers $k > 2$.

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

Adding $\|v_{k+1}\|$ where $v_{k+1} \in \mathbb{R}^n$ is some arbitrary vector in $\mathbb{R}^n$ to both sides will not change the truth value of the inequality giving us

$$\|v_1 + v_2 + \dots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| + \|v_{k+1}\|$$

Using the triangle inequality again we can get a second inequality involving the left hand side:

$$\|v_1 + v_2 + \cdots + v_k + v_{k+1}\| \leq \|v_1 + v_2 + \cdots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \cdots + \|v_k\| + \|v_{k+1}\|$$

Due to the transitive properties of inequalities we get that:

$$\|v_1 + v_2 + \cdots + v_k + v_{k+1}\| \leq \|v_1\| + \|v_2\| + \cdots + \|v_k\| + \|v_{k+1}\|$$

Which is the statement $P(k+1)$ is so we have shown that if $P(k)$ is true then $P(k+1)$ is true.

So we have shown that for $k > 2$, $P(k) \Rightarrow P(k+1)$ and that $P(1)$ and $P(2)$ are true. Then by the method of induction we have shown that $P(k)$ is true for all $k \in \mathbb{Z}_+$.

Last Part

$$\|v_1 + v_2 + \cdots + v_k\| = \|v_1\| + \|v_2\| + \cdots + \|v_k\|$$

is true if $\forall i, j \in [1, k], \exists k \in \mathbb{R}_+$ such that $v_i = k \cdot v_j$.